Hypothesis testing

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Overview

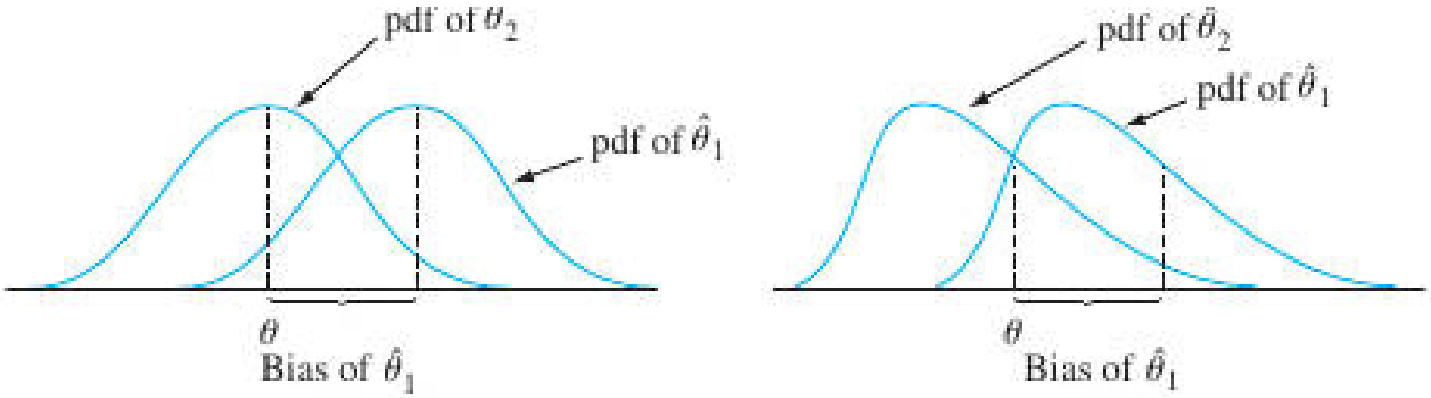
1. Point Estimation
2. Preliminaries
3. Confidence Interval
4. Hypothesis Testing
5. p-values

Point Estimation (1)

- From **sample parameters** mean $\bar{x}$ and variance s^2, **population parameters** μ and σ^2 can be **inferred** or estimated
- θ is the parameter to be estimated/ inferred.
- **Definition:** A **point estimate** of a parameter θ is a single number that gives a sensible value for θ that is obtained by selecting a suitable statistic and computation of its value from sample data. The selected statistic is called the **point estimator** of θ
- **Example:** Find point estimate of population variance σ^2, given sample data $x = \{44.2, 43, 9, 44.7, 44.2, 44.0, 43.8, 44.6, 43.1\}$ from a population distribution.
- **Solution:** $\hat{\sigma}^2 = \frac{\sum_{i=1}^{8}(x_i - \bar{x})^2}{n-1} = 0.25125$; Thus, $\hat{\sigma} = 0.501$
- It is possible to find estimator $\hat{\theta}$ for which $\hat{\theta} = \theta$, ie. $\hat{\theta} = \theta +$ error of estimation, minimize $(\hat{\theta} - \theta)^2$, ie. Mean squared error $MSE = E((\hat{\theta} - \theta)^2)$, difficult to find smallest MSE

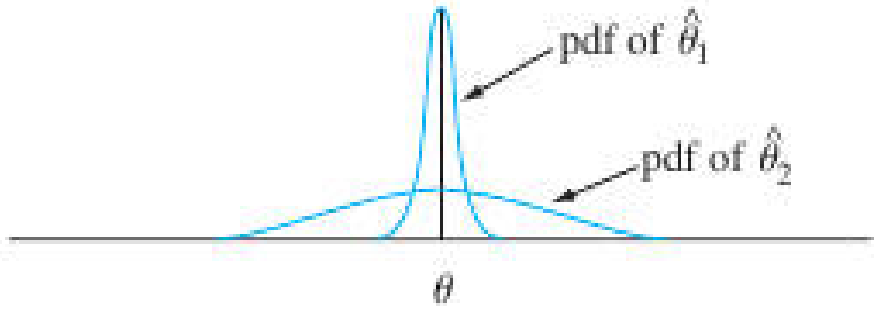
Point Estimation (2)

- **Definition:** A point estimator $\hat{\theta}$ is said to be **unbiased edtimator** of θ if $E(\hat{\theta}) = \theta$ for every possibe value of θ. If $\hat{\theta}$ is **not unbiased**, $E(\hat{\theta}) - \theta$ is the **bias** of $\hat{\theta}$
- pdf of **biased estimator** $\hat{\theta}_1$ and **unbiased estimator** $\hat{\theta}_2$ for a parameter θ
- **Example: sample proportion** $\hat{p} = \frac{X}{n}$ is an unbiased estimator, where p is probability of successes in binomial distribution with parameters n and p. $E(\hat{\theta}) = E(\hat{p}) = E(\frac{X}{n}) = \frac{1}{n}E(X) = \frac{1}{n}np = p = \theta$, since binomial distribution is normal distribution with $E(X) = np$



Some examples of point estimator

- **Proposition:** Let $X_1, X_2, \cdots, X_n$ be the random sample from a distribution with μ and variance σ^2, then the estimator $\hat{\sigma}^2 = s^2 = \frac{\sum(X_i - \bar{(X)})^2}{n-1}$ is an **unbiased** estimator of σ^2,**but** $\hat{s}$ is **biased estimator** of σ
- **Proposition:** If $X_1, X_2, \cdots, X_n$ are the random sample from a distribution with μ and variance σ^2, then $\bar{X}$ is an **unbiased** estimator of μ
- **Estimator with minimum variance:** Among all estimators of θ that are unbiased, choose one that has minimum variance. Such $\hat{\theta}$ is called **Minimum Variance Unbiased Estimator**(MVUE). For $X_1, X_2, \cdots, X_n$ random sample from Normal distribution with μ and σ^2, $\hat{\mu} = \bar{X}$ is MVUE for μ, $\hat{\theta}_1$ is MVUE



Standard error of estimator

- **Standard error(SE) of an estimator** $\hat{\theta}$ is standard deviation $\sigma_\theta = \sqrt{Var(\hat{\theta})}$; for example Standard Error of Mean (SEM) is $\sigma/\sqrt{n}$
- If SE has unknow parameter, SE needs to be estimated , then $\hat{\sigma}_\theta$ is the **estimated SE**
- **Example:** SE of $\hat{p} = \frac{X}{n}$ is

$$\sigma_{\hat{p}} = \sqrt{Var(\frac{X}{n})} = \sqrt{\frac{Var(X)}{n^2}} = \sqrt{\frac{npq}{n^2}} = \sqrt{\frac{pq}{n}}$$

- Since p is unknown, substitute $\hat{p} = X/n$ and $\hat{q} = 1 - X/n$ in $\sigma_{\hat{p}}$ to obtain $\hat{\sigma}_{\hat{p}} = \sqrt{\frac{\hat{p}\hat{q}}{n}}$
- If $\hat{\theta}$ has **approximately normal distribution** for large n, typically, **true value** of θ lies within 2 SE or Standard deviation of $\hat{\theta}$
- Point estimates are obtained by bootstrap method or maximum likelihood estimator method

Normal Distribution and Standard Normal Distribution

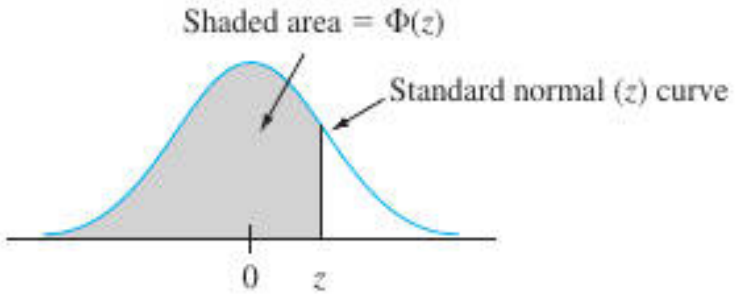
- A continuous rv X is said to have a **normal distribution** with parameters μ and σ (or μ and σ^2), where $-\infty < \mu < \infty$ and $0 < \sigma$, if the PDF of X is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \tag{1}$$

- The **normal distribution** with parameters $\mu = 0$ and $\sigma = 1$ is called **standard normal distribution**. A **random variable** having standard normal distribution is called a **standard normal variable**, denoted by Z. $f(z; 0, 1)$ is standard normal or z curve; the pdf of Z is

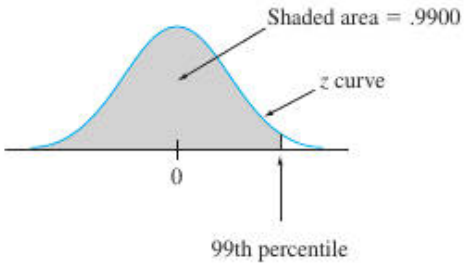
$$f(z; 0, 1) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{-z^2}{2}}; -\infty < z < \infty \tag{2}$$

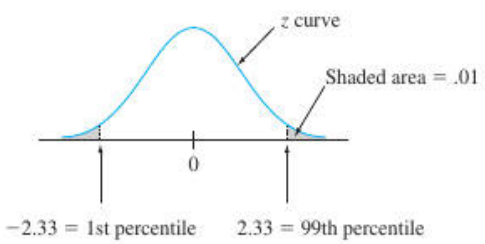
- The cdf of Z is $P(Z \leq z) = \int\limits_{-infty}^{z} f(y; 0, 1)dy = \phi(z)$, where $z = \frac{x-\mu}{\sigma}$



Percentile of Standard Normal Distribution

- 99th **percentile** of Std. Normal is that value on x axis such that area under z curve to the left of the value is 0.99, ie, for which value of z $P(Z \leq z) = 0.99$? the $100p^{th}$ percentile is row and column of normal distribution cdf table in which the entry p is found.
- z_α notation: For statistical inference we need value on z axis for which α of the area under z curve lies to the right of z_α, z_α is called z **critical value**





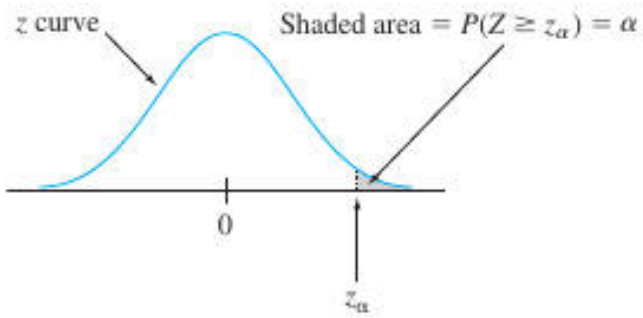


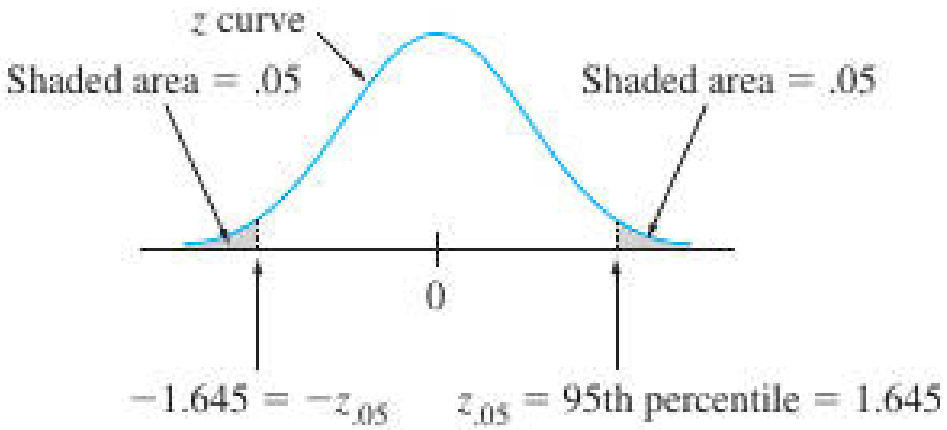
Figure 4.17 Finding the 99th percentile

Table 4.1 Standard Normal Percentiles and Critical Values

Percentile	90	95	97.5	99	99.5	99.9	99.95
α (tail area)	.1	.05	.025	.01	.005	.001	.0005
$z_\alpha = 100(1-\alpha)$th percentile	1.28	1.645	1.96	2.33	2.58	3.08	3.27

Percentile of Standard Normal Distribution: Example

$z_{.05}$ is the $100(1 - .05)$th $= 95$th percentile of the standard normal distribution, so $z_{.05} = 1.645$. The area under the standard normal curve to the left of $-z_{.05}$ is also .05. (See Figure 4.20.)



Percentile of Non-Standard Normal Distribution

If X has a normal distribution with mean μ and standard deviation σ, then

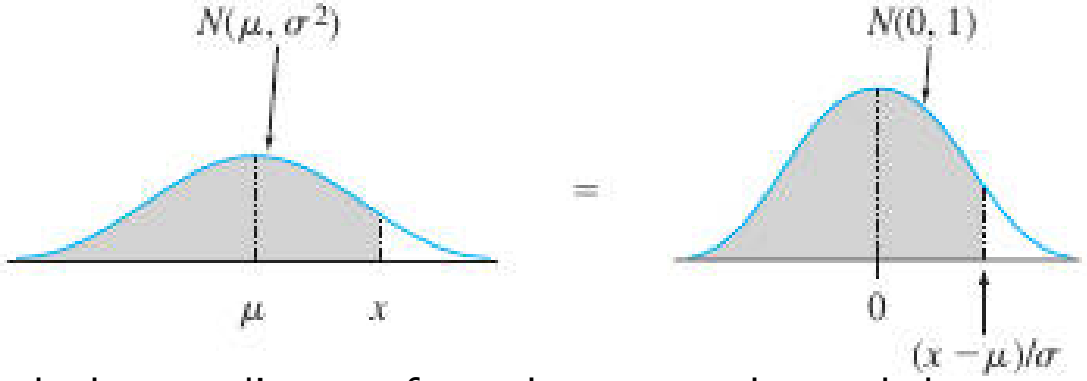
$$Z = \frac{X - \mu}{\sigma}$$

has a standard normal distribution. Thus

$$P(a \le X \le b) = P\left(\frac{a-\mu}{\sigma} \le Z \le \frac{b-\mu}{\sigma}\right)$$

$$= \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)$$

$$P(X \le a) = \Phi\left(\frac{a-\mu}{\sigma}\right) \qquad P(X \ge b) = 1 - \Phi\left(\frac{b-\mu}{\sigma}\right)$$



- Standardizing calculates a distance from the mean value and then reexpresses the distance as some number of standard deviations. Thus, if $\mu = 100$ and $s = 15$, then $x = 130$ corresponds to $z = (130 - 100)/15 = 30/15 = 2$. That is, 130 is 2 standard deviations above (to the right of) the mean value.

Empirical rule

- If the population distribution of a variable is (approximately) normal, then
  - Roughly 68% of the values are within 1 SD of the mean.
  - Roughly 95% of the values are within 2 SDs of the mean.
  - Roughly 99.7% of the values are within 3 SDs of the mean.
- $(100p)^{th}$ percentile for normal $(\mu, \sigma) = \mu + (100p)^{th}$ percentile for standard normal . σ

Statistical Interval based on Single Sample

- Point estimate states nothing about how close an estimator $\hat{\theta}$ will be to the true θ
- **Confidence Interval** (CI) gives an interval around the true θ with a **confidence level** (measures degree of reliability of interval)
- A **Confidence level** of 95% means 95% of all samples would give an interval that includes true θ and only 5% of would yield erroneous interval; frequently used **Confidence levels** are 95%, 99%, 90%
- if **confidence level is high** and **interval is narrow**, it implies knowledge of parameter is **reasonably precise**, **wide CI** gives uncertainity

Basic properties of CI (1)

- Let the parameter of interest be population mean μ, population distribution is normal and value of the population **standard deviation** σ is known
- The actual sample observations $x_1, x_2, \cdots, x_n$ are assumed to be the result of a random sample $X_1, \cdots, X_n$ from a normal distribution with mean value μ and standard deviation σ
- The sample mean $\bar{X}$ is normally distributed with expected value μ and standard deviation $\sigma/\sqrt{n}$. Standardizing X gives standard normal variable $Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$
- Because the area under the standard normal curve between -1.96 and 1.96 is .95,

$$P(-1.96 < \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < 1.96) = 0.95$$

- Write above as $l < \mu < u$, where the endpoints l and u involve X and $\sigma/\sqrt{n}$.

Basic properties of CI (2)

1. Multiply through by $\sigma/\sqrt{n}$:

$$-1.96 \cdot \frac{\sigma}{\sqrt{n}} < \overline{X} - \mu < 1.96 \cdot \frac{\sigma}{\sqrt{n}}$$

2. Subtract $\overline{X}$ from each term:

$$-\overline{X} - 1.96 \cdot \frac{\sigma}{\sqrt{n}} < -\mu < -\overline{X} + 1.96 \cdot \frac{\sigma}{\sqrt{n}}$$

3. Multiply through by -1 to eliminate the minus sign in front of μ (which reverses the direction of each inequality):

$$\overline{X} + 1.96 \cdot \frac{\sigma}{\sqrt{n}} > \mu > \overline{X} - 1.96 \cdot \frac{\sigma}{\sqrt{n}}$$

that is,

$$\overline{X} - 1.96 \cdot \frac{\sigma}{\sqrt{n}} < \mu < \overline{X} + 1.96 \cdot \frac{\sigma}{\sqrt{n}}$$

The equivalence of each set of inequalities to the original set implies that

$$P\left(\overline{X} - 1.96\frac{\sigma}{\sqrt{n}} < \mu < \overline{X} + 1.96\frac{\sigma}{\sqrt{n}}\right) = .95$$

- random quantity appears on two ends, unknown constant μ appears in the middle. In interval notation, $\left(\bar{X} - 1.96.\sigma/\sqrt{n}, \bar{X} + 1.96.\sigma/\sqrt{n}\right)$
- The probability is .95 that the random interval includes or covers the true value of μ

Confidence Interval: Definition

If, after observing $X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n$, we compute the observed sample mean $\bar{x}$ and then substitute $\bar{x}$ into (7.4) in place of $\bar{X}$, the resulting fixed interval is called a **95% confidence interval for μ**. This CI can be expressed either as

$$\left(\bar{x} - 1.96 \cdot \frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96 \cdot \frac{\sigma}{\sqrt{n}}\right) \text{ is a 95\% CI for } \mu$$

or as

$$\bar{x} - 1.96 \cdot \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + 1.96 \cdot \frac{\sigma}{\sqrt{n}} \text{ with 95\% confidence}$$

A concise expression for the interval is $\bar{x} \pm 1.96 \cdot \sigma/\sqrt{n}$, where $-$ gives the left endpoint (lower limit) and $+$ gives the right endpoint (upper limit).

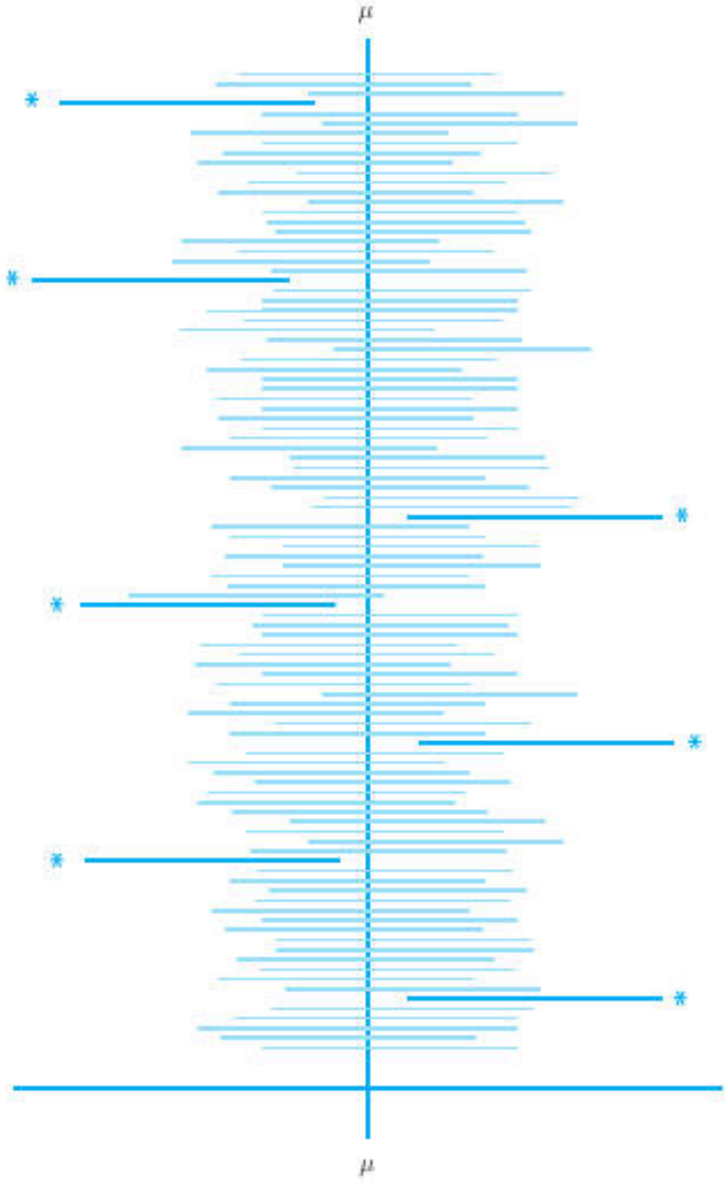
The quantities needed for computation of the 95% CI for true average preferred height are $\sigma = 2.0$, $n = 31$, and $\bar{x} = 80.0$. The resulting interval is

$$\bar{x} \pm 1.96 \cdot \frac{\sigma}{\sqrt{n}} = 80.0 \pm (1.96)\frac{2.0}{\sqrt{31}} = 80.0 \pm .7 = (79.3, 80.7)$$

That is, we can be highly confident, at the 95% confidence level, that $79.3 < \mu < 80.7$. This interval is relatively narrow, indicating that μ has been rather precisely estimated. ■

Confidence Interval: Interpretation

- An event A has probability .95 implies if the experiment on which A is defined is performed over and over again, in the long run A will occur 95% of the time
- CI is **not** $P(\mu \text{ lies in} (79.3, 80.7)) = .95$
- Let A be the event that $\bar{X} - 1.96.\sigma/\sqrt{n} < \mu < \bar{X} + 1.96.\sigma/\sqrt{n}$. Since $P(A) = .95$, in the long run (in repeated experiments on different samples) 95% of computed CIs will contain μ.



Other Confidence Interval: Example

A probability of $(1-\alpha)$ is achieved by using $z_{\alpha/2}$ in place of 1.96.

CI is point estimate of $\mu \pm$(z critical value).(standard error of the mean)

A **100(1 − α)% confidence interval** for the mean μ of a normal population when the value of σ is known is given by

$$\left(\bar{x} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}\right) \tag{7.5}$$

or, equivalently, by $\bar{x} \pm z_{\alpha/2} \cdot \sigma/\sqrt{n}$.

The production process for engine control housing units of a particular type has recently been modified. Prior to this modification, historical data had suggested that the distribution of hole diameters for bushings on the housings was normal with a standard deviation of .100 mm. It is believed that the modification has not affected the shape of the distribution or the standard deviation, but that the value of the mean diameter may have changed. A sample of 40 housing units is selected and hole diameter is determined for each one, resulting in a sample mean diameter of 5.426 mm. Let's calculate a confidence interval for true average hole diameter using a confidence level of 90%. This requires that $100(1-\alpha) = 90$, from which $\alpha = .10$ and $z_{\alpha/2} = z_{.05} = 1.645$ (corresponding to a cumulative z-curve area of .9500). The desired interval is then

$$5.426 \pm (1.645)\frac{.100}{\sqrt{40}} = 5.426 \pm .026 = (5.400, 5.452)$$

With a reasonably high degree of confidence, we can say that $5.400 < \mu < 5.452$. This interval is rather narrow because of the small amount of variability in hole diameter ($\sigma = .100$).

One sided CI

A **large-sample upper confidence bound for μ** is

$$\mu < \bar{x} + z_\alpha \cdot \frac{s}{\sqrt{n}}$$

and a **large-sample lower confidence bound for μ** is

$$\mu > \bar{x} - z_\alpha \cdot \frac{s}{\sqrt{n}}$$

A **one-sided confidence bound for p** results from replacing $z_{\alpha/2}$ by z_α and $\pm$ by either $+$ or $-$ in the CI formula (7.10) for p. In all cases the confidence level is approximately $100(1-\alpha)\%$.

The slant shear test is the most widely accepted procedure for assessing the quality of a bond between a repair material and its concrete substrate. The article "Testing the Bond Between Repair Materials and Concrete Substrate" (*ACI Materials J.*, 1996: 553–558) reported that in one particular investigation, a sample of 48 shear strength observations gave a sample mean strength of 17.17 N/mm^2 and a sample standard deviation of 3.28 N/mm^2. A lower confidence bound for true average shear strength μ with confidence level 95% is

$$17.17 - (1.645)\frac{(3.28)}{\sqrt{48}} = 17.17 - .78 = 16.39$$

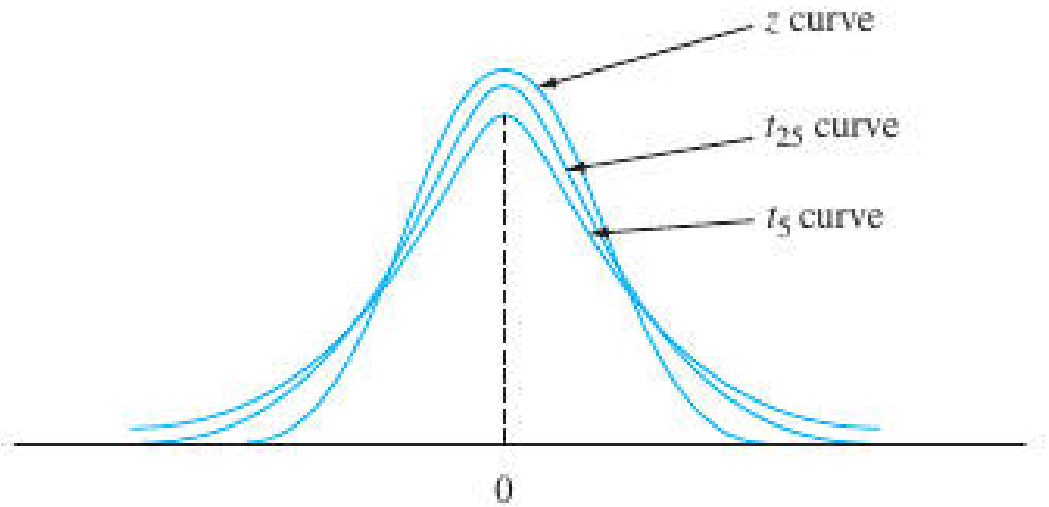
That is, with a confidence level of 95%, we can say that $\mu > 16.39$.

Normal population distribution: t distribution

- **Assumption:** The population of interest is normal, so that $X_1, \cdots, X_n$ constitutes a random sample from a normal distribution with both μ and σ unknown.
- For large n, the random variable $Z = (\bar{X} - \mu)/(S/\sqrt{n})$ has approximately a **standard normal distribution**. When n is small, S is no longer likely to be close to σ, so the variability in the distribution of Z arises from randomness in both the numerator and the denominator.
- This implies that the probability distribution of $Z = (\bar{X} - \mu)/(S/\sqrt{n})$ will be more spread out than the standard normal distribution. This introduces a new family of probability distributions called t-distributions.
- **Definition:** When $\bar{X}$ is the mean of a random sample of size n from a normal distribution with mean μ, the rv $T = (\bar{X} - \mu)/(S/\sqrt{n})$ has a probability distribution called a t-distribution with $n - 1$ degrees of freedom (df).

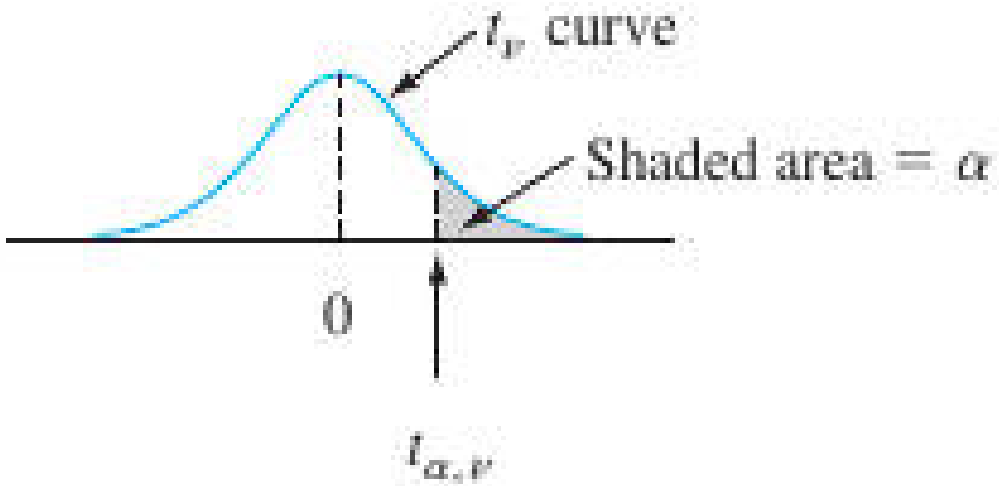
Properties of t distribution (1)

- T **does not have standard normal distribution**, when n is **small**.
- Any particular t distribution results from specifying the value of a single parameter, called the **number of degrees of freedom (df)**, ν; different curves for different ν values. Let t_ν denote the t-distribution with ν df
- Each t_ν curve is bell-shaped and centered at 0.
- Each t_ν curve is more spread out than the standard normal (z) curve.
- As ν increases, the spread of the corresponding t_ν curve decreases.
- As $n \to \infty$, the sequence of t_ν curves approaches the standard normal curve (so the z curve is often called the t curve with df $= \infty$).



Properties of t distribution (2)

- The number of df ν for T $n-1$ because, although S is based on the n deviations $X_1 - \bar{X}, \cdots, X_n - \bar{X}$, $\sum(X_i - \bar{X}) = 0$ implies that only $n-1$ of these are **freely determined.** The number of df for a t variable is the number of **freely determined deviations** on which the estimated standard deviation in the denominator of T is based.
- t distribution is used for making inferences along with ν by capturing **t-curve tail areas** like z_α for z curve.
- **t critical value:** Let $t_{\alpha,n}$ is the number on the measurement axis for which the area under the t curve with n df to the right of $t_{\alpha,n}$ is α; $t_{\alpha,n}$ is called a t critical value.



One sample t CI

- The standardized variable T has a t distribution with $n-1$ df, and the area under the t density curve between $-t_{\alpha/2,n-1}$ and $t_{\alpha/2,n-1}$ is $1-\alpha$ (area $\alpha/2$ lies in each tail), so $P(-t_{\alpha/2,n-1} < T < t_{\alpha/2,n-1}) = 1-\alpha$
- **Proposition:**

Let $\bar{x}$ and s be the sample mean and sample standard deviation computed from the results of a random sample from a normal population with mean μ. Then a **$100(1-\alpha)\%$ confidence interval for μ** is

$$\left(\bar{x} - t_{\alpha/2,n-1} \cdot \frac{s}{\sqrt{n}},\ \bar{x} + t_{\alpha/2,n-1} \cdot \frac{s}{\sqrt{n}}\right) \tag{7.15}$$

or, more compactly, $\bar{x} \pm t_{\alpha/2,n-1} \cdot s/\sqrt{n}$.

An **upper confidence bound for μ** is

$$\bar{x} + t_{\alpha,n-1} \cdot \frac{s}{\sqrt{n}}$$

and replacing $+$ by $-$ in this latter expression gives a **lower confidence bound for μ,** both with confidence level $100(1-\alpha)\%$.

One sample t CI: Example

- Let there is a sample data X with $n = 30$ observations. Determine the mean μ of the population with 95% confidence level.
- Let $\bar{x}$ of sample data is 1203.191 and sample $s = 543.54$.
- The CI is based on $n - 1 = 29$ degrees of freedom, so the necessary t **critical value** is $t_{.025,29} = 2.045$.
- The interval estimate is $\mp t_{.025,29}.s/\sqrt{n} = (7000.253, 7406.129)$
- $7000.253 < \mu < 7406.129$ with 95% confidence.
- A **lower** 95% confidence bound would result from retaining only the lower confidence limit (the one with $-$ sign) and replacing 2.045 with $t_{.05,29} = 1.699$.
- When sample size $n < 40$, apply t distribution rather than z or standard normal.

Hypothesis and Test procedure

There is a need for justifing a claim using data when there are contradictory claims about parameters estimated.

- A **statistical hypothesis**, or just hypothesis, is a **claim** or assertion either about the **value of a single parameter** (population characteristic or characteristic of a probability distribution), about the values of everal parameters, or about the form of an entire probability distribution.
- Statistical hypothesis is formed such that the **claim** is initially **favoured** and is not rejected until and unless sample evidence contradicts it and provide strong support in favour of alternative claim.
- **Null Hupothesis** H_0 is the claim that is initially assumed to be **true** (the prior belief claim); Null typically means- of no value, no change or consequences etc.
- **Alternative Hypothesis** H_a is the assertion that is **contradictory** to H_0
- **Hypothesis testing** is the procedure of finding evidence from the data either to **reject** H_0 or **fail to reject** H_0

Hypothesis Testing: Example

- **Problem:** There is 10% defective items(p) during manufacturing in a particular process. We want to know, for a different process, whether the rate of defects reduces or remains same.
- $H_0 : p = 0.10$, $H_a = p < 0.10$
- H_0 is typically of the form $H_0 : \theta = \theta_0$, where θ is any parameter of interest and θ_0 is some constant. Three forms of H_a typically:
  - $H_a : \theta \neq \theta_0$
  - $H_a : \theta < \theta_0$
  - $H_a : \theta > \theta_0$

Test Procedure: Exmaple

- Above problem; random sample of $n = 200$, X : number of defective items in sample, A **binomial** random variable represents observed value of X.
- if H_0 is true, $E(X) = np$, since binomial rv. So, $E(X) = 200(0.10) = 20$; ie. we can expect fewer than 20 defective items if H_a is true. So, it is reasonable to reject H_0 only if x is substantially less than 20.

Hypothesis Testing Procedure

- A **test statistic** or function of sample data is neded to make decision
- A **rejection region** consisting of those x values for which H_0 will be rejected in favour of H_a
- For previous example, **test statistic** is $E(X)$, and **rejection region** is any value from 0 to 15 (substantially lower than 20)
- If observed /computed **test statistic value** falls in **rejection region**, H_0 is **rejected**

Thus, A test procedure is specified by the following:

- A **test statistic**, a function of the sample data on which the decision (reject H_0 or do not reject H_0) is to be based
- A **rejection region**, the set of all test statistic values for which H_0 will be rejected
- The null hypothesis will then be rejected if and only if the observed or computed test statistic value falls in the rejection region.

Errors in Hypothesis testing

- Consider a rejection region $x \leq 15$ (a cut-off) for defective item problem. It might happen that an unusual sample results in $x = 13$, even when H_0 is true, then H_0 is erroneously rejected, ie. **Type I error**
- When H_a is true, an unusual sample might yield $x = 20$, in which case H_0 will not be rejected; **type II error**
- Sampling variability generates these error due to unrepresentattive sample
- **Definition:** A **type I error** consists of **rejecting the null hypothesis** H_0 when H_0 **is true**. A **type II error** involves **not rejecting** H_0 when H_0 **is false**.
- A good procedure is one for which the probability of making either type of error is small.
- **The choice of a particular rejection region cutoff value fixes the probabilities of type I and type II errors.**
- These error probabilities are traditionally denoted by α and β, respectively. Because H_0 specifies a unique value of the parameter, there is a single value of α. However, there is a different value of β for each value of the parameter consistent with H_a.

Errors in Hypothesis testing: Example 1 (1)

Car testing: No visible damage at 25% of time in 10 km/hour crash test. A modified bumper design is proposed to increase percentage. p : proportion of all 10 km/hour crashes with new bumper that results in no visible damage. Let $n = 20$ for new design. Compute the errors of hypothesis testing.

- $H_0 : p = 0.25$ (no improvement); $H_a : p > 0.25$
- H_0 should be rejected if substantial number of crashes show no damage.
- **Test statistic:** X : number of crashes with no visible damage; x observed value of test statistic
- **Rejection region:** $R_8 = \{8, 9, \cdots, 20$, if $x \geq 8$, in rejection region, reject H_0. This rejection region is called **upper-tailed** because it consists only of large values of the test statistic.
- When H_0 is true, X has binomial probability distribution with $n = 20, p = 0.25$. Then

$$\begin{aligned}\alpha &= P(\text{type I error}) = P(H_0 \text{is rejected} | H_0 \text{ is true}) \\ &= P(X \geq 8 \text{when} X \sim Bin(20, 0.25)) \\ &= 1 - B(7; 20, 0.25) = 1 - 0.898 = 0.102 \sim 10\%\end{aligned}$$

Errors in Hypothesis testing: Example 1 (2)

- That means when H_0 is actually **true**, roughly 10% of all experiments consisting of 20 crashes would result in H_0 being **incorrectly rejected** (a type I error).
- But, there are **different possible** values of *beta*, as $p = 0.3, 0.5, \cdots$ and many more. For example, $\beta(0.3) =$

$$= P(\text{type II error when p=0.3}) = P(H_0 \text{is not rejected} | H_0 \text{is false because} p = 0.$$
$$= P(X < 7 \text{when} X \sim Bin(20, 0.25))$$
$$= B(7; 20, 0.25) = 0.772 \sim 77\%$$

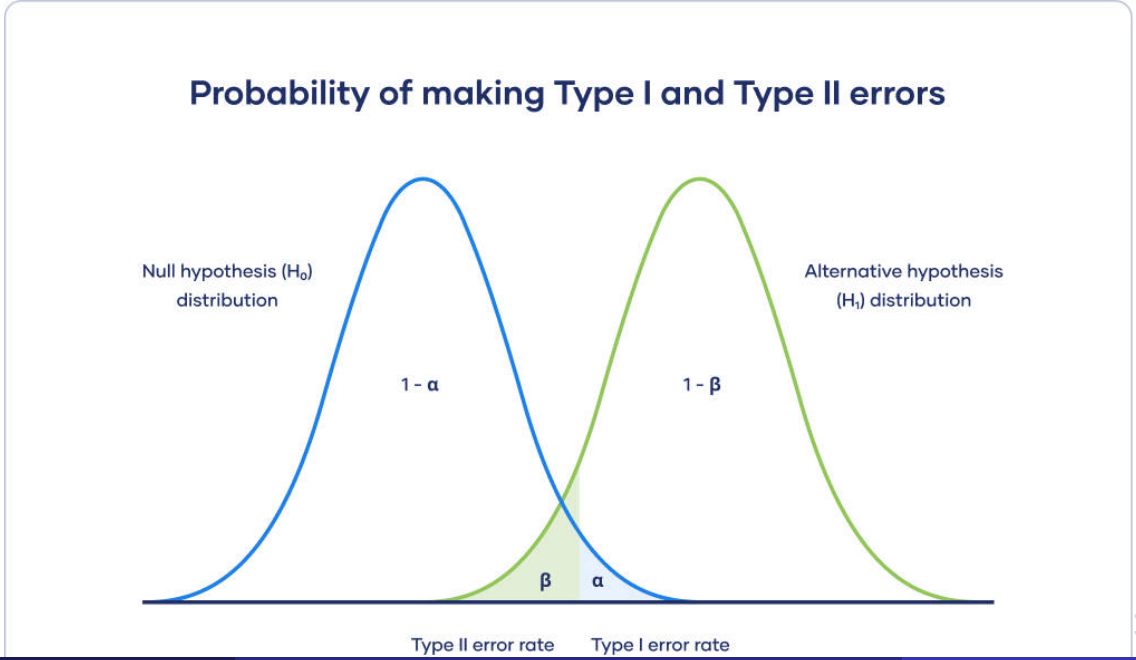
- When p is actually .3 rather than .25 (a "small" departure from H_0), roughly 77% of all experiments of this type would result in H_0 being **incorrectly not rejected**!

The accompanying table displays β for selected values of p (each calculated for the rejection region R_8). Clearly, β decreases as the value of p moves farther to the right of the null value .25. Intuitively, the greater the departure from H_0, the less likely it is that such a departure will not be detected.

p	.3	.4	.5	.6	.7	.8
$\beta(p)$	.772	.416	.132	.021	.001	.000

Errors in Hypothesis testing: Example 1 (3)

- The proposed test procedure is still reasonable for testing the more realistic null hypothesis that $p \leq .25$.
- This means many different α possible for different p's, ie. α for each p that is at most .25: $\alpha(.25), \alpha(.23), \alpha(.20)$ and so on.
- However for any such $p < .25$, $\alpha(p) < \alpha(.25) = .102$. That is, the largest value of α occurs for the boundary value .25 between H_0 and H_a. Thus if α is small for the simplified null hypothesis, it will also be as small as or smaller for the more realistic H_0.



Errors in Hypothesis testing (4)

- **Proposition:** Suppose an experiment and a sample size are fixed and a test statistic is chosen. Then **decreasing the size of the rejection region** to obtain a **smaller value** of α results in a larger value of β for any particular parameter value consistent with H_a.
- Since type I error is more serious, specify largest value of α that can be tolerated and find a rejection region having that value of α. Thus, β becomes as small as possible
- This α is called **significance level**, typically $0.10, 0.05, 0.01$
- The test procedure is called level α test (level .05 test), i.e, A test with significance level α is one for which the **type I error** probability is controlled at the specified level.

Errors in Hypothesis testing Example (4)

- Let μ is the **true average nicotine content** of brand B cigarettes. The objective is to test $H_0 : \mu = 1.5$ versus $H_a : \mu > 1.5$ based on a random sample $X_1, X_2, \cdots, X_{32}$ of nicotine content.
- Suppose the distribution of nicotine content is known to be **normal** with $\sigma = 0.20$. Then $\bar{X}$ is **normally distributed** with mean value $\mu_{\bar{X}} = \mu$ and standard deviation $\sigma_{\bar{X}} = 0.20/\sqrt{32} = 0.0354$. Assume H_0 is true.
- Test statistic: standardized $\bar{X}$,$Z = \frac{\bar{X}-1.5}{\sigma/\sqrt{n}}$
- Z expresses the distance between $\bar{X}$ and its expected value when H_0 is **true** as some number of standard deviations. $z = 3$ results from an $\bar{x}$ that is 3 standard deviations **larger** than we would have expected it to be when H_0 is true.
- Rejecting H_0 when $\bar{x}$ "considerably" exceeds 1.5 is equivalent to rejecting H_0 when z "considerably" exceeds 0. That is, the form of the rejection region is $z \geq c$.
- Determine c so that $\alpha = .05$.
- When H_0 is true, Z has a standard normal distribution. Thus,

Errors in Hypothesis testing Example (5)

$$\alpha = P(\text{type I error}) = P(\text{rejecting } H_0 \text{ when } H_0 \text{ is true})$$
$$= P(Z \geq c \text{ when } Z \approx N(0,1))$$

- The value c must capture upper-tail area .05 under the z curve. From Table, $c = z_{.05} = 1.645$.
- Notice that $z \geq 1.645$ is equivalent to $\bar{x} - 1.5 \geq (.0354)(1.645)$, that is, $\bar{x} \geq 1.56$. Then β involves the probability that $\bar{X} < 1.56$ and can be calculated for any μ greater than 1.5.

Test about population mean: Case I & II

Case I & II : Normal population with known σ and Unknown population with unknown σ (Large Sample Test) uses Z statistic
case III: population distribution is at least approximately normal (n is small), uses T statistic

- μ has a value μ_0. Let $X_1, \cdots, X_n$ represent a random sample of size n from the normal population. Sample mean $\bar{X}$ has a normal distribution with expected value $\mu_{\bar{X}} = \mu$ and standard deviation $\sigma_{\bar{X}} = \sigma/\sqrt{n}$.
- When H_0 is **true**, $\mu_{\bar{X}=\mu_0}$.
- Consider Z statistic obtained by standardizing $\bar{X}$ under the assumption that H_0 is true, $Z = \frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}}$
- Substitution of the computed sample mean $\bar{x}$ gives z, the distance between observed $\bar{x}$ and μ_0 expressed in "standard deviation units.".
- Example: $H_0 : \mu = 100$, $\sigma_{\bar{X}} = \sigma/\sqrt{n} = 2.0$, and $\bar{x} = 103$, then the test statistic value is $z = (103 - 100)/2.0 = 1.5$. That is, the observed value of $\bar{x}$ is 1.5σ (of $\bar{X}$) larger than what we expect it to be when H_0 is true.
- If this distance is **too large** in a direction consistent with H_a, H_0 should be **rejected**.

Test about population mean: Case I & II (1)

a) $H_a : \mu > \mu_0$**:**

- The rejection region is $z \geq c$, i.e, $z \geq z_\alpha$, the cutoff value c should be chosen to control the probability of a type I error at the desired level α
- The required cutoff c is the z critical value that captures **upper-tail** area α under the z curve. The rejection region $z \geq z_\alpha$ has type I error probability α. The test procedure is **upper-tailed** because the rejection region consists only of large values of the test statistic.
- **Example:** Let $c = 1.645$, i.e, $z_{.05} = 1.645$, i.e $\alpha = .05$

$$\alpha = P(\text{type I error}) = P(H_0 \text{is rejected when } H_0 \text{ is true})$$

$$= P(Z \geq 1.645 \text{ when } Z \approx N(0,1)) = 1 - \Phi(1.645) = 0.5$$

Test about population mean: Case I & II (2)

b) $H_a : \mu < \mu_0$**:**

- The rejection region is $z \leq -c$, i.e, $z \leq -z_\alpha$, the cutoff value c should be chosen to control the probability of a type I error at the desired level α
- The required cutoff $-c$ is the z critical value that captures **lower-tail** area α under the z curve. The rejection region $z \leq -z_\alpha$ has type I error probability α. The test procedure is **lower-tailed** because the rejection region consists only of smaller values of the test statistic.
- **Example:** Let $c = -1.28$, i.e, $-z_{.10} = -1.28$, i.e $\alpha = .10$

$$\alpha = P(\text{type I error}) = P(H_0 \text{is rejected when } H_0 \text{ is true})$$

$$= P(Z \leq -1.28 \text{ when } Z \approx N(0,1)) = \Phi(-1.28) = 0.10$$

Test about population mean: Case I & II (3)

c) $H_a : \mu \neq \mu_0$**:**

- H_0 should be rejected if $\bar{X}$ is too far from **either side of** μ_0, i.e, the rejection region is $z \geq c$ or $z \leq -c$.
- **Example:** Let $\alpha = .05$

$$.05 = P(\text{type I error}) = P(H_0 \text{is rejected when } H_0 \text{ is true})$$

$$= P(Z \geq c \text{ or } Z \leq -c \text{ when } Z \approx N(0,1)) = 1 - \Phi(c) + \Phi(-c) = 2(1 - \Phi(c))$$

- c is such that the area under z curve to the right of c is 0.025, i.e, $c = 1.96$, i.e, $z_{.025} = 1.96$ and rejection region is $z \geq 1.96$ or $z \leq -1.96$

Summary: $H_0 = \mu = \mu_0$, Test Statistic $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$, Rejection Region for level α test

- $H_a : \mu > \mu_0 : z \geq z_\alpha$; upper tailed test
- $H_a : \mu < \mu_0 : z \leq -z_\alpha$; lower tailed test
- $H_a : \mu \neq \mu_0 : z \geq z_{\alpha/2}$ or $z \leq -z_{\alpha/2}$; two tailed test

Test about population mean: Case I & II (4)

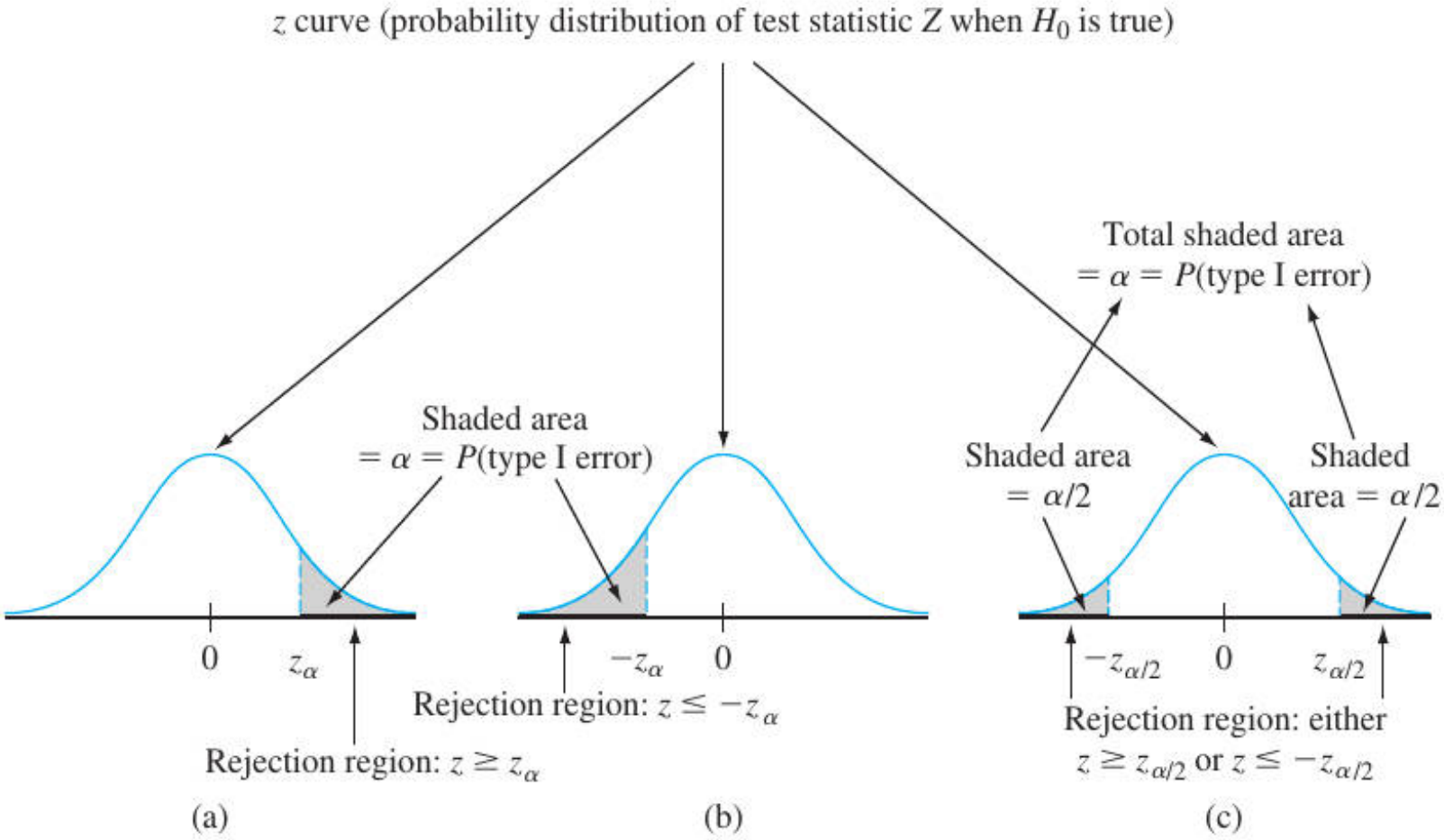


Figure 8.2 Rejection regions for z tests: (a) upper-tailed test; (b) lower-tailed test; (c) two-tailed test

Steps of Hypothesis Testing

1. Identify the parameter of interest and describe it in the context of the problem situation.
2. Determine the null value and state the null hypothesis.
3. State the appropriate alternative hypothesis.
4. Give the formula for the computed value of the test statistic (substituting the null value and the known values of any other parameters, but *not* those of any sample-based quantities).
5. State the rejection region for the selected significance level α.
6. Compute any necessary sample quantities, substitute into the formula for the test statistic value, and compute that value.
7. Decide whether H_0 should be rejected, and state this conclusion in the problem context.

Steps of Hypothesis Testing: Example (1)

A manufacturer of sprinkler systems used for fire protection in office buildings claims that the true average system-activation temperature is 130°. A sample of $n = 9$ systems, when tested, yields a sample average activation temperature of 131.08°F. If the distribution of activation times is normal with standard deviation 1.5°F, does the data contradict the manufacturer's claim at significance level $\alpha = .01$?

1. Parameter of interest: μ = true average activation temperature.
2. Null hypothesis: H_0: $\mu = 130$ (null value $= \mu_0 = 130$).
3. Alternative hypothesis: H_a: $\mu \neq 130$ (a departure from the claimed value in *either* direction is of concern).
4. Test statistic value:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{\bar{x} - 130}{1.5/\sqrt{n}}$$

Steps of Hypothesis Testing: Example (2)

5. Rejection region: The form of H_a implies use of a two-tailed test with rejection region *either* $z \geq z_{.005}$ *or* $z \leq -z_{.005}$. From Section 4.3 or Appendix Table A.3, $z_{.005} = 2.58$, so we reject H_0 if either $z \geq 2.58$ or $z \leq -2.58$.

6. Substituting $n = 9$ and $\bar{x} = 131.08$,

$$z = \frac{131.08 - 130}{1.5/\sqrt{9}} = \frac{1.08}{.5} = 2.16$$

That is, the observed sample mean is a bit more than 2 standard deviations above what would have been expected were H_0 true.

7. The computed value $z = 2.16$ does not fall in the rejection region ($-2.58 < 2.16 < 2.58$), so H_0 cannot be rejected at significance level .01. The data does not give strong support to the claim that the true average differs from the design value of 130.

Test about population mean: Case III

If $X_1, X_2, \cdots, X_n$ is a random sample from an approximately normal distribution for small n, where σ is unknown, the standardized variable $T = \frac{\bar{X}-\mu}{S\ \sqrt{n}}$ has t-distribution with $n-1$ degrees of freedom.

The One-Sample t Test

Null hypothesis: H_0: $\mu = \mu_0$

Test statistic value: $t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$

Alternative Hypothesis	Rejection Region for a Level α Test
H_a: $\mu > \mu_0$	$t \geq t_{\alpha,n-1}$ (upper-tailed)
H_a: $\mu < \mu_0$	$t \leq -t_{\alpha,n-1}$ (lower-tailed)
H_a: $\mu \neq \mu_0$	either $t \geq t_{\alpha/2,n-1}$ or $t \leq -t_{\alpha/2,n-1}$ (two-tailed)

p-value

- Another way of reaching a conclusion in a hypothesis testing analysis by computation of a probability value called p-value
- P-value provides an intuitive measure of the **strength of evidence in the data against** H_0.
- **Definition:** The P-value is the probability, calculated assuming that the **null hypothesis is true**, of obtaining a value of the test statistic **at least as contradictory to** H_0 **as the value calculated from the available sample**.

  - The P-value is a probability.
  - This probability is calculated assuming that the null hypothesis is true.
  - Beware: The P-value is not the probability that H_0 is true, nor is it an error probability!
  - To determine the P-value, we must first decide which values of the test statistic are at least as contradictory to H_0 as the value obtained from our sample.

p-value : Example (1)

A sample of 51 Panasonic AAA batteries gave a sample mean $\bar{x}$ zinc mass of 2.06 g and a sample standard deviation s of 0.141 g. Does this data provide compelling evidence for concluding that the population mean μ zinc mass exceeds 2.0 g?

- $H_0 : \mu = 2.0$; $H_a : \mu > 2.0$
- Z statistic: $z = \frac{2.06-2.0}{0.141/\sqrt{51}} = 3.04$
- Decide which values of z are as contradictory to H_0
- **First attempt:** Which values of $\bar{x}$ are at least as contradictory to the null hypothesis as 2.06? Because $>$ appears in H_a, 2.10 is at least as contradictory to H_0 as is 2.06, and so in fact is any $\bar{x}$ value that exceeds 2.06.
- But an $\bar{x}$ value that exceeds 2.06 corresponds to a value of z that exceeds 3.04. Thus
  $P - value = P(Z \geq 3.04 \text{ when } \mu = 2.0) = 1 - \Phi(3.04) = .0012$

P-value Rule

- P-value = .250: In this case, fully 25% of all possible test statistic values are at least as contradictory to H_0 as the one that came out of our sample. So our data is not all that contradictory to the null hypothesis.
- P-value = .0018: Here, only .18% (much less than 1%) of all possible test statistic values are at least as contradictory to H_0 as what we obtained. Thus the sample appears to be highly contradictory to the null hypothesis.

- The **smaller** the P-value, the more evidence there is in the sample data **against the null hypothesis** and for the alternative hypothesis.
- That is, H_0 should be **rejected** in favor of H_a when the P-value is **sufficiently small**. So what constitutes "sufficiently small"?

Decision rule based on the P-value

Select a significance level α (as before, the desired type I error probability). Then

$$\text{reject } H_0 \text{ if } P\text{-value} \leq \alpha$$
$$\text{do not reject } H_0 \text{ if } P\text{-value} > \alpha$$

P-value Rule: Exmaple

- **Example 1:** Calculated $P - value = .0012$. For significance level $\alpha = .01$, reject H_0 because $.0012 < .01$. However, for $\alpha = .001$, which requires more substantial evidence from the data before H_0 can be rejected; we would **not reject** H_0 because $.0012 > .001$.
- **Example 2:** Let $H_0 : \mu = 1.5$; $H_a : \mu > 1.5$, let $z = 2.10$; $P - value = 1 - \Phi(2.10) = .0179$

$$\alpha = .10 \Rightarrow z_\alpha = z_{.10} = 1.28 \Rightarrow 2.10 \geq 1.28 \Rightarrow \text{reject } H_0$$
$$\alpha = .05 \Rightarrow z_\alpha = z_{.05} = 1.645 \Rightarrow 2.10 \geq 1.645 \Rightarrow \text{reject } H_0$$
$$\alpha = .01 \Rightarrow z_\alpha = z_{.01} = 2.33 \Rightarrow 2.10 < 2.33 \Rightarrow \text{do not reject } H_0$$

- **Proposition:** The P-value is the smallest significance level α at which the null hypothesis can be rejected. Because of this, the P-value is alternatively referred to as the observed significance level (OSL) for the data.

P-value Rule and α : Interpretation

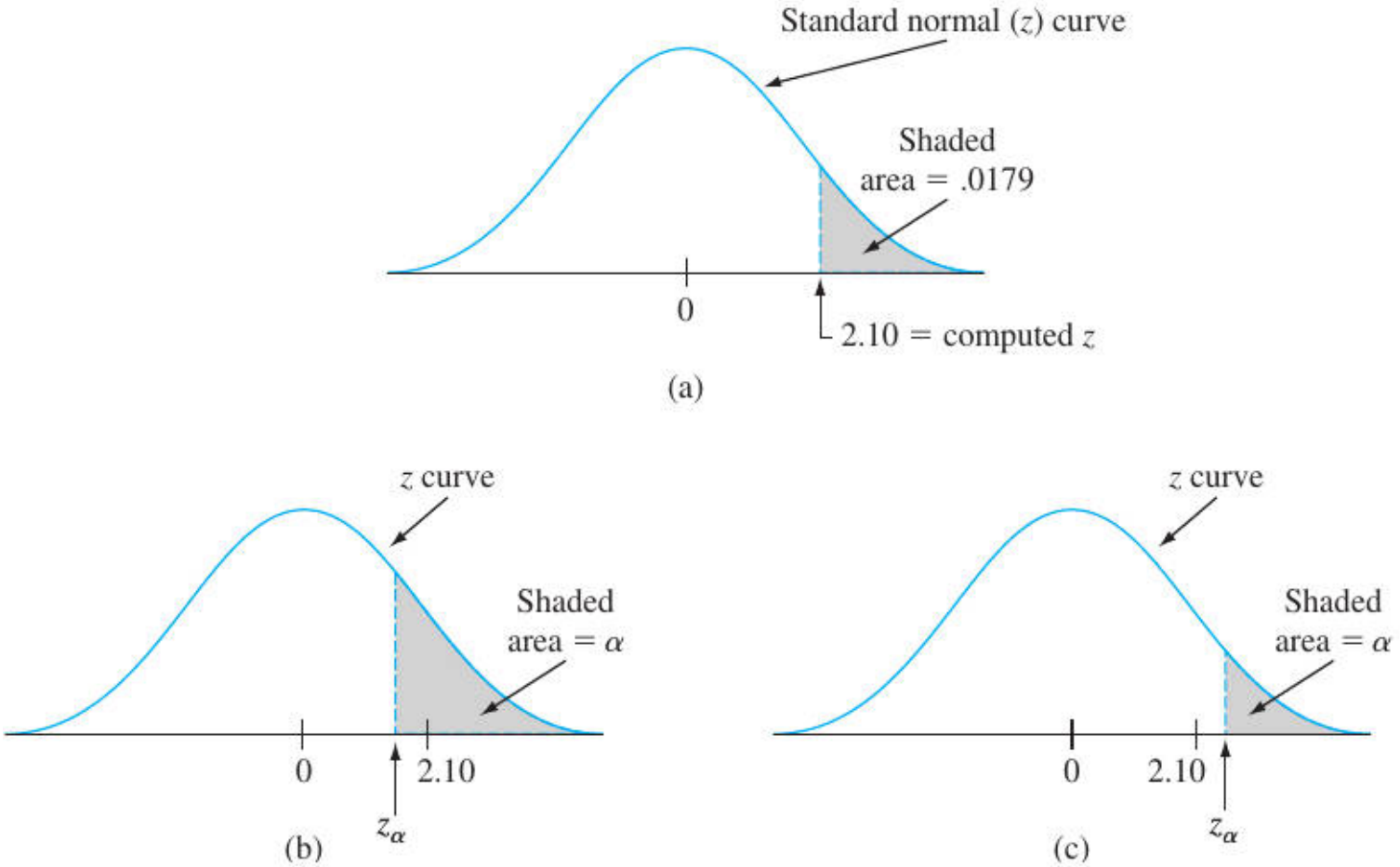


Figure 8.7 Relationship between α and tail area captured by computed z: (a) tail area captured by computed z; (b) when $\alpha < .0179$, $z_\alpha < 2.10$ and H_0 is rejected; (c) when $\alpha < .0179$, $z_\alpha > 2.10$ and H_0 is not rejected

Note: Figure caption: (b) when $\alpha > .0179$, $z_\alpha < 2.10$ and H_0 is rejected.

P-value and α

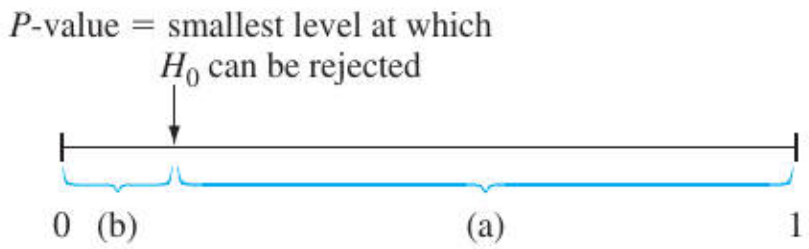


Figure 8.8 Comparing α and the P-value: (a) reject H_0 when α lies here; (b) do not reject H_0 when α lies here

- **Note 1:** The calculation of the P-value depends on whether the test is upper-, lower-, or two-tailed. However, once it has been calculated, the comparison with α does not depend on which type of test was used.
- **Note 2:** With the P-value one can see which significance levels H_0 would or would not be rejected and thus can choose their own significance level. Further, knowing the P-value helps to distinguish between a close call (e.g., $\alpha = .05, P - value = .0498$) and a very clearcut conclusion (e.g., $\alpha = .05, P - value = .0003$).

P-values for z Test

$$P\text{-value: } P = \begin{cases} 1 - \Phi(z) & \text{for an upper-tailed } z \text{ test} \\ \Phi(z) & \text{or an lower-tailed } z \text{ test} \\ 2[1 - \Phi(|z|)] & \text{for a two-tailed } z \text{ test} \end{cases}$$

- **Note 1:** Each of these is the probability of getting a value at least as extreme as what was obtained (assuming H_0 true)

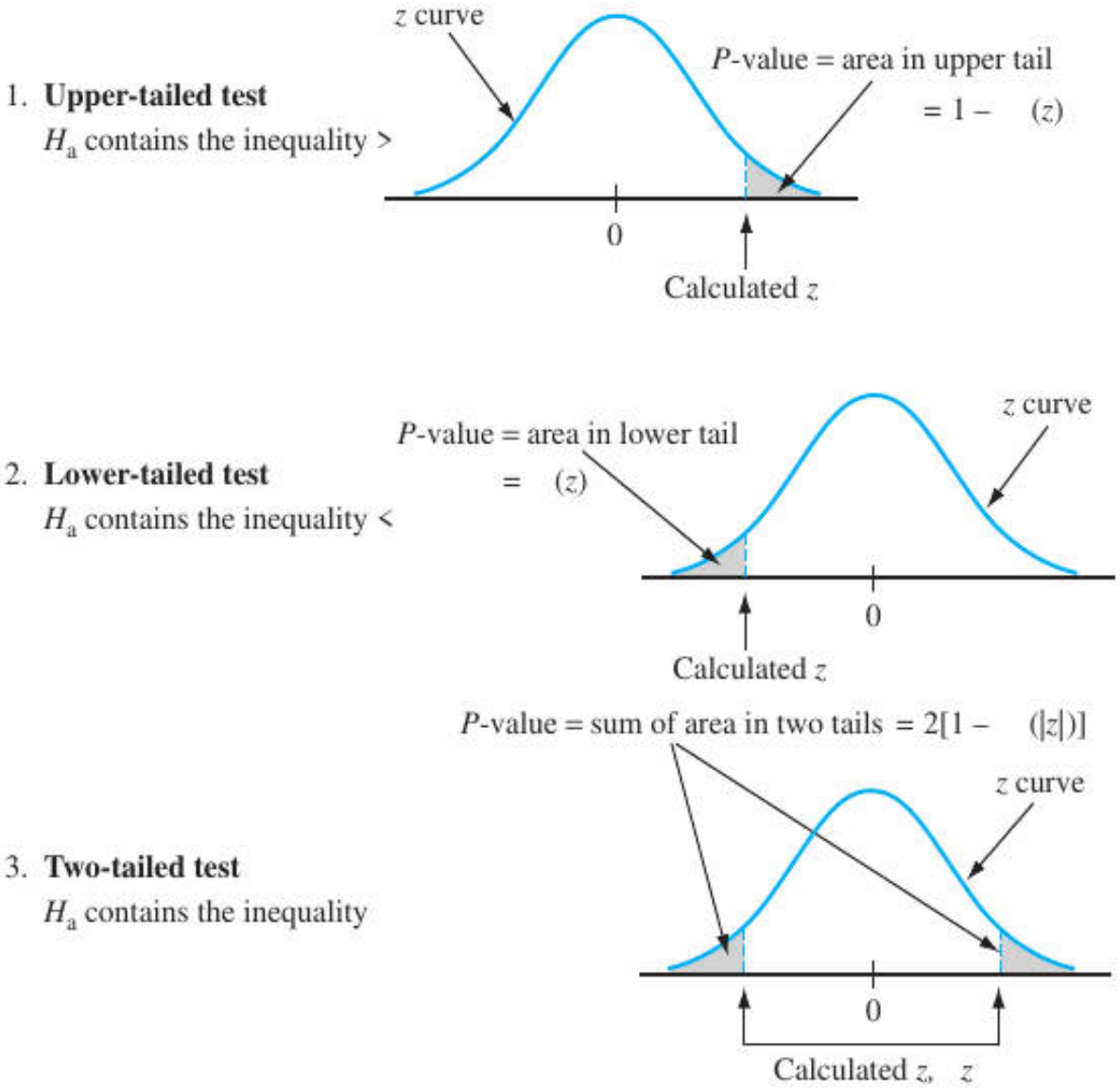


Figure 8.9 Determination of the P-value for a z test

P-values for z Test : Example

The target thickness for silicon wafers used in a certain type of integrated circuit is 245 μm. A sample of 50 wafers is obtained and the thickness of each one is determined, resulting in a sample mean thickness of 246.18 μm and a sample standard deviation of 3.60 μm. Does this data suggest that true average wafer thickness is something other than the target value?

1. Parameter of interest: μ = true average wafer thickness
2. Null hypothesis: H_0: $\mu = 245$
3. Alternative hypothesis: H_a: $\mu \neq 245$
4. Formula for test statistic value: $z = \dfrac{\bar{x} - 245}{s/\sqrt{n}}$
5. Calculation of test statistic value: $z = \dfrac{246.18 - 245}{3.60/\sqrt{50}} = 2.32$
6. Determination of P-value: Because the test is two-tailed,

$$P\text{-value} = 2(1 - \Phi(2.32)) = .0204$$

7. Conclusion: Using a significance level of .01, H_0 would not be rejected since $.0204 > .01$. At this significance level, there is insufficient evidence to conclude that true average thickness differs from the target value. ■

P-values for t Test

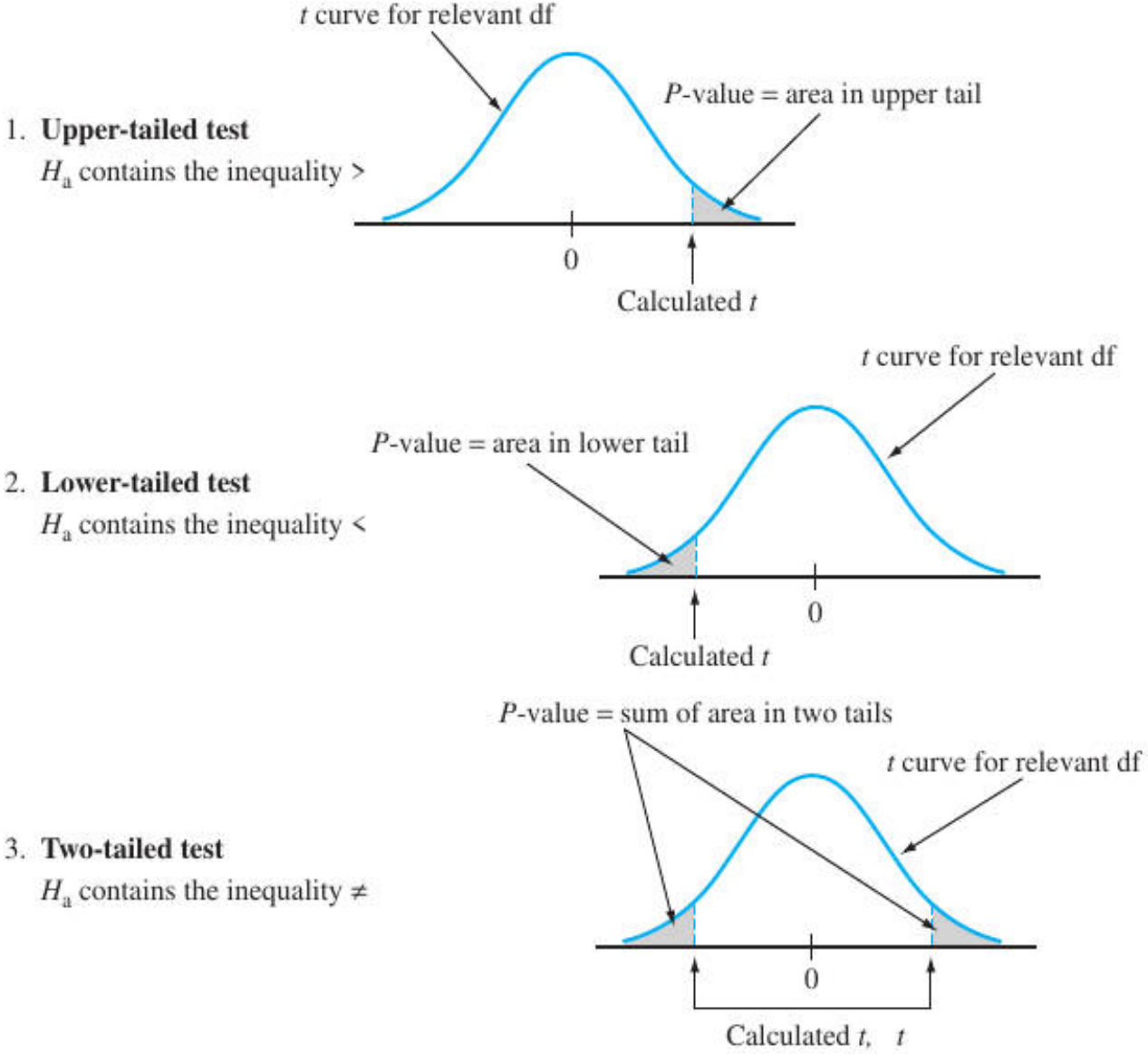


Figure 8.10 *P*-values for t tests